

Preliminary Implementation Caveats & Engineering Audit

Technical Audit & Architectural Fidelity Assessment | July 2026

Executive Summary & Portfolio Framing: This repository represents an **Engineering-Complete ETL & QA Framework Prototype**. The core ETL cleaning pipeline (Rules 1–4), spatial coordinate transformations, PopulationSim IPU utilities, and 6-Gate QA Suite math are genuinely implemented to specification. However, the downstream causal core—including GTFS-RT Causal Proxy 2 delay matching, the DTA feedback loop, and the upper-level nested logit choice coefficients—are currently implemented as functional architectural scaffolding utilizing synthetic/illustrative constants pending empirical Maximum Likelihood Estimation (MLE) calibration.

1. Code Reality vs. Specification Audit Matrix

The table below compares the Version 7.0 specification requirements against the actual Python code reality in backend/ and executive-engine/.

Component	Spec Requirement	Code Reality & Implementation Status	Status Assessment
DataSF Ingestion	Paginate with \$limit=50000 + offset loop	ingestion.py hardcodes \$limit=1000 without an offset loop. Silently truncates windows with >1,000 records.	Truncated Stub
NHTSA SGO	Pull bulk SGO CSV releases (VIN, automation category)	Calls general CrashAPI/GetCaseList endpoint returning empty mock list (wrong API endpoint).	Non-functional Stub
GTFS-RT Feed	Persistent poller building trip update time series	Fully mocked/hardcoded with {'data': []}; zero live API calls executed.	Mocked Stub
Rule 4 (CRS)	Reproject WGS84 to EPSG:26910 prior to metric joins	Implemented strictly as specified. Executed first in run_etl_pipeline().	Implemented to Spec
Rule 1 (Dedup)	Composite key: VIN + ±5min + 50m spatial buffer + union-find	Fully implemented via GeoPandas sjoin and NetworkX union-find connected components.	Implemented to Spec
Rules 2 & 3	SF boundary filter & GTFS speed anomaly ceilings (65mph/500m/s)	Implemented as specified in cleaning.py.	Implemented to Spec
PUMA-TAZ Crosswalk	Block-level population-weighted areal apportionment	Areal weighting logic is fully written, but falls back to a 3-row mock crosswalk when raw shapefiles are absent.	Code Ready (Fallback Mock)
IPU Convergence	<0.0001 relative weight change or 100-iter cap	Implemented correctly in populationsim_utils.py.	Implemented to Spec
QA Gates 1–6	6 hard-threshold validation gates with MAPE/RMSE constraints	All 6 gates implemented returning (passed, statistic) tuples with exact spec thresholds.	Implemented to Spec
Nested Logit Utilities	Empirically estimated mode choice coefficients (b_0_AV, b_fare)	Contains explicit code comments # Dummy coefficients. Constants are hand-set, not estimated.	Dummy Constants
DTA Feedback Loop	RMSE-based skim convergence feedback loop	Loop uses rmse_skim *= 0.5 each iteration—artificial math convergence, not driven by skim comparison.	Artificial Convergence
Proxy 2 (Transit Delay)	Spatial (50m) + temporal (±15min) incident-to-GTFS matching	Code comment admits: 'Here we mock the merged result for demonstration'—delays are random uniform() draws.	Synthetic Demonstration
Bayesian SPRT	ODD-stratified informative prior (Road Class × Speed × Time)	Prior is a single flat human incident rate rather than ODD-stratified.	Simplified Prior

2. Soundness & Interpretation of Dashboard Results

There are no empirical policy findings present in the current release. The dashboards (React Operational Dashboard & Streamlit Executive Engine) function as **interactive method prototypes** and structural demonstrations rather than validated estimates:

- **Executive Engine Simulation:** Corridor metrics (Market St, Mission St, Geary Blvd, etc.) respond dynamically to slider controls based on illustrative response parameters (e.g., base_vmt, base_dh_ratio, corridor tax sensitivity) defined directly within app.py, rather than live backend pipeline outputs.
- **Nested Logit Modal Shift:** Downstream calculations (transit shift %, consumer surplus, VOT by income tier) inherit dummy logit coefficients. Consequently, these metrics demonstrate the analytical mechanics of the engine, not calibrated behavioral phenomena.
- **QA Gate Behavior on Synthetic Data:** Gates 1–6 pass cleanly on synthetic/mock inputs by construction (e.g., Gate 1 returns (True, set()) when inputs are empty). Passing QA suites confirm software execution integrity, not empirical validity of AV impacts.

3. Specific Technical Caveats & Methodological Limits

ID	Caveat Title	Technical Description & Methodological Impact
C-1	Placebo Independence (Gate 4)	Adjacent San Francisco corridors share underlying spatial congestion shocks. Without spatial-clustering standard error corrections, the $p > 0.10$ threshold is overly permissive, inflating confidence in causal delay estimates.
C-2	Uncalibrated Logit Utilities	Logit parameters (b_0_AV, b_fare, comfort discounts) are hardcoded dummy values. Every equity and VOT calculation is conditional on constants not yet fitted via MLE.
C-3	IPU Cell Sparsity at Gate 6	Even with the $n \geq 30$ cell count floor, passing Gate 6 only confirms absence of miscalibration in well-populated TAZ×income cells, not population-wide demographic accuracy.
C-4	Specification Version Labeling	The core specification text internally labels itself '6.3-PRODUCTION' while external files reference 'Version 7.0'. Audit trails should maintain distinction between spec versions.
C-5	Non-Stratified Control Window	Proxy 2's 21-day baseline control window does not incorporate day-of-week or holiday stratification in proxy_pipeline.py, creating potential bias if control spans non-representative periods.
GOV	VMT Denominator Enforcement	While the specification dictates using CPUC commercial filings exclusively for VMT denominators (excluding DMV testing VMT), no code-level validation currently prevents accidental data blending.

Conclusion for Audit & Interview Contexts: Frame this codebase as an 'Engineering-complete ETL and QA infrastructure prototype with stubbed analytical layers pending empirical calibration' rather than a system generating finalized policy estimates.